

Fu Chen (First-year CMU SCS Master Student)

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EDUCATION

Carnegie Mellon University, United States Aug 2026 - Dec 2027 (expected)
M.S. in Computational Data Science (MCDS) at LTI, School of Computer Science GPA: 4.33/4.33

The Chinese University of Hong Kong, Shenzhen, China Sept 2022 - Jul 2026
B.E. in Computer Science and Engineering (Artificial Intelligence Stream) Cumulative GPA: 3.85/4.00 (top 3%)

- a. **Excellent student** [Math] Deep/Reinforcement Learning, Probabilities & Statistics, Advanced Algebra;
(Major GPA 4.00/4.00) [Computer] C/C++/Java/Python Programming, Algorithm Analyses (Graduate), etc.
b. **Experienced teacher** Teaching Fellow of Linear Algebra, Discrete Mathematics, and Computer Architecture.
c. **Award recipient** Academic Performance Scholarship & Dean's List $\times 3$, and Student Teaching Fellow Award.

***National University of Singapore**, Singapore May 2024 - Dec 2024
Exchange Student, Research Assistant *via UROPS Program* GPA: 5.00/5.00

***University of California, Los Angeles** - Extension, United States Jan 2025 - Apr 2025
Visiting Student *via Study Abroad Program* GPA: 4.00/4.00

PUBLICATIONS

- Chen, F.**, Tian, L., Fei, T., & Yu, T.* (2024). Fast-scBatch: Batch effect correction using neural network-driven distance matrix adjustment. *Proceedings of the 15th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics*, 1–6.
- Liu, X.*[#], **Chen, F.**[#], Pan, J., Ai, C., Guo, F.*, & Tang, J.* (2025). MediHerb: A multi-modal enhanced framework for disease inference via herbal knowledge. *Computational Biomedicine*, 1:202509.

[#] indicates co-first authorship; * indicates the corresponding author(s)

ACADEMIC EXPERIENCE

Faster Machine Learning Methods to Solve Batch Effects | Prof. Tianwei Yu @CUHK Jul 2023 - May 2024

- Improved deep learning models to eliminate batch effects, enabling faster downstream analyses of single-cell data.
- Applied dimensionality reduction to lower time complexity and random sampling for further improvement.

Computational Methods for Phylogenetic Trees Analyses | Prof. Louxin Zhang @NUS May 2024 - Dec 2024

- Utilized vector representations to analyze phylogenetic trees, including tree operations and visualization.
- Developed machine learning methods to identify common substructures among multiple phylogenetic trees and assess their similarity; published clear, visually appealing visualization software.

Statistical Methods for Identifying Cell Topology | Prof. Jingyi Jessica Li @UCLA Sept 2024 - Jul 2025

- Identified cell structures and topologies to infer potential cellular motility from single-cell RNA data.
- Applied statistical methods to test hypotheses and validated findings with empirical simulations.

Attention-based Model for TCM Herb-Disease Inference | Independent Study Jul 2025 - Oct 2025

- Developed an attention-based multimodal model for herb-disease inference in Traditional Chinese Medicine.
- Achieved state-of-the-art performance in herb inference; built a webpage for dataset and model exploration.

Sandboxed Execution of High-Risk OpenClaw Medical Skills | Independent Study Feb 2026 - Mar 2026

- Collected and developed medical-related skills and open-sourced them on GitHub, achieving 1.2k+ stars in 7 days; evaluated biomedical paper reproduction using local OpenClaw; Focused on privacy and safety in medical scenarios

INDUSTRY EXPERIENCE

MLE Intern | Alibaba Inc. May 2026 - Aug 2026
SWE Intern | Microsoft Jan 2026 - Mar 2026

COMPETITIONS

National Olympiad in Informatics of China | Bronze Medalist Jul 2021
Asia and Pacific Informatics Olympiad | Silver Medalist Aug 2020

SKILLS

Language Chinese (native); English (fluent; TOEFL-107)

Software Proficient in MS Excel, VS Code, R Studio, Cursor, Conda/uv, Git, Markdown/L^AT_EX editors, Tableau

Programming Proficient in Python (PyTorch), C/C++, Java, R and MySQL; Familiar with Matlab and VHDL